

Lab4: Design and implement C/C++ Program to find shortest paths from a given vertex in a weighted connected graph to other vertices using Dijkstra's algorithm

```
#include<stdio.h>
void dij();
int i, j, n, visited[20], source, cost[20][20], d[20];
void main()
{

printf("Enter no. of vertices: ");
scanf("%d", &n);
printf("Enter the cost matrix (999 for self loop and no edge)\n");
for (i = 1; i <= n; i++)
{
for (j = 1; j <= n; j++)
{
scanf("%d", &cost[i][j]);
}
}
printf("\nEnter the source node: ");
scanf("%d", &source);
dij();
for (i = 1; i <= n; i++)
{
if (i != source)
printf("\nShortest path from %d to %d is %d", source, i, d[i]);
}
}
void dij()
{
int i, j, min, u, w;
for (i = 1; i <= n; i++)
{
visited[i] = 0;
d[i] = cost[source][i];
}
visited[source] = 1;
d[source] = 0;
for (j = 2; j <= n; j++)
{
min = 999;
for (i = 1; i <= n; i++)
{
if(!visited[i])
{
```

```

        if (d[i] < min)
        {
min = d[i];
u = i;
        }
    }
    visited[u] = 1;
    for (w = 1; w <= n; w++)
    {
        if (cost[u][w] != 999 && visited[w] == 0)
        {
            if(d[w] > cost[u][w] + d[u])
d[w] = cost[u][w] + d[u];
        }
    }
}
}
}
}

```

OUTPUT

Enter no. of vertices: 5

Enter the cost matrix (999 for self loop and no edge)

999 3 999 7 999

3 999 4 2 999

999 4 999 5 6

7 2 5 999 4

999 999 6 4 999

Enter the source node: 1

Shortest path from 1 to 2 is 3

Shortest path from 1 to 3 is 7

Shortest path from 1 to 4 is 5

Shortest path from 1 to 5 is 9